RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Saturday, 12 September 2026, 2:04 PM
State	Finished
Completed on	Saturday, 12 September 2026, 2:05 PM
Time taken	1 min 11 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Divide and Conquer function to find the first occurrence of 0
int findFirstZero(int arr[], int low, int high) {
    if (high >= low) {
        int mid = low + (high - low) / 2;

        // Check if mid is the first 0
        if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0)
            return mid;

        // If the element is 1, the first 0 must be in the right half
        if (arr[mid] == 1)
            return findFirstZero(arr, mid + 1, high);

        // If the element is 0 but not the first one, it's in the left half
        return findFirstZero(arr, low, mid - 1);
    }
    return -1; // Return -1 if no 0 is found
}

int main() {
    int m;
    if (scanf("%d", &m) != 1) return 0;

    int arr[m];
    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    int firstZeroIdx = findFirstZero(arr, 0, m - 1);

    // If no zero is found, count is 0; otherwise, subtract index from size
    int zeroCount = (firstZeroIdx == -1) ? 0 : (m - firstZeroIdx);

    printf("%d\n", zeroCount);
    return 0;
}

```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔

	Input	Expected	Got	
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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